Financial hardship in Amsterdam

Studentnames and studentnumbers here

2026-06-22

Set-up your environment

```
f{r package_install, include=FALSE} # install.packages("tidyverse") # install.packages("yaml")
# install.packages("rmarkdown") # install.packages("ggplot2") # install.packages("dplyr")
# install.packages(c("sf", "tmap" , "cbsodataR")) # install.packages("tinytex") #tinytex::install_tinyt
```

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Tutorial 1 Group 4

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Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Social Problem: People in Amsterdam who struggle to make ends meet.

Many people living in Amsterdam have difficulty covering their daily expenses, such as rent, food and energy bills. This financial pressure can lead to stress, debt, and a lower quality of life.

Why is this relevant?

This issue has become increasingly relevant due to rising housing costs, inflation, and unequal income growth across households. Recent data from the Centraal Bureau voor de Statistiek (CBS, 2025) show that in 2024, 13.3% of Dutch adults reported having difficulty making ends meet, while financial dissatisfaction remains concentrated among lower-income groups. Additionally, 26.1% of adults reported having many financial worries about their future, indicating that financial insecurity remains a structural issue in Dutch society.

Research by Mullainathan and Shafir (2013) shows that financial scarcity can lead to chronic stress, reduced decision-making capacity, and lower overall wellbeing. In cities like Amsterdam, where living costs and rent prices have increased substantially over the last two decades, lower-income households are more vulnerable to financial pressure.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

Limitations of the datasets:

This study has several limitations.

First, the datasets are used at neighbourhood level, which means individual household variation cannot be observed. This may lead to a problem where conclusions of individuals are based on group-level data

Second, the main variable (*Moeite met rondkomen*) is data made from self reports, meaning that subjective interpretation may affect responses.

Third, the student variable may overestimate the actual student population, since not everyone aged 15–25 is enrolled in education.

Finally, important variables such as debt, savings, household composition, education level, and employment type are missing. These factors may also strongly influence financial situations.

2.3 Describe the type of variables included

Part 3 - Quantifying

3.1 Data cleaning

Before analysis, the datasets required several cleaning steps to make sure that the graphs could be made without gaps and with consistent data.

First, to start off after importing the raw datasets, all datasets were filtered to include only neighbourhood-level observations for Amsterdam. This improved the focus of the analysis on Amsterdam.

Second, irrelevant variables such as identification numbers, age categories, and margin-of-error variables were removed to simplify the dataset.

Third, string variables such as neighbourhood names and codes were cleaned using `trimws()` to remove unnecessary spaces. This was necessary for merging datasets accurately.

Fourth, duplicate observations were removed using `distinct()`, ensuring that each neighbourhood appeared only once in the dataset. While this improves consistency, it may hide meaningful repeated measurements.

Lastly, missing values were excluded during calculations by applying `na.rm = TRUE`. This prevents computational errors.

To sum the impact of these changes up, the steps improved overall data quality and usefulness, however they did reduce some detail and have potential limitations.

3.2 Generate necessary variables

To improve the analysis, we made new variables from the original datasets.

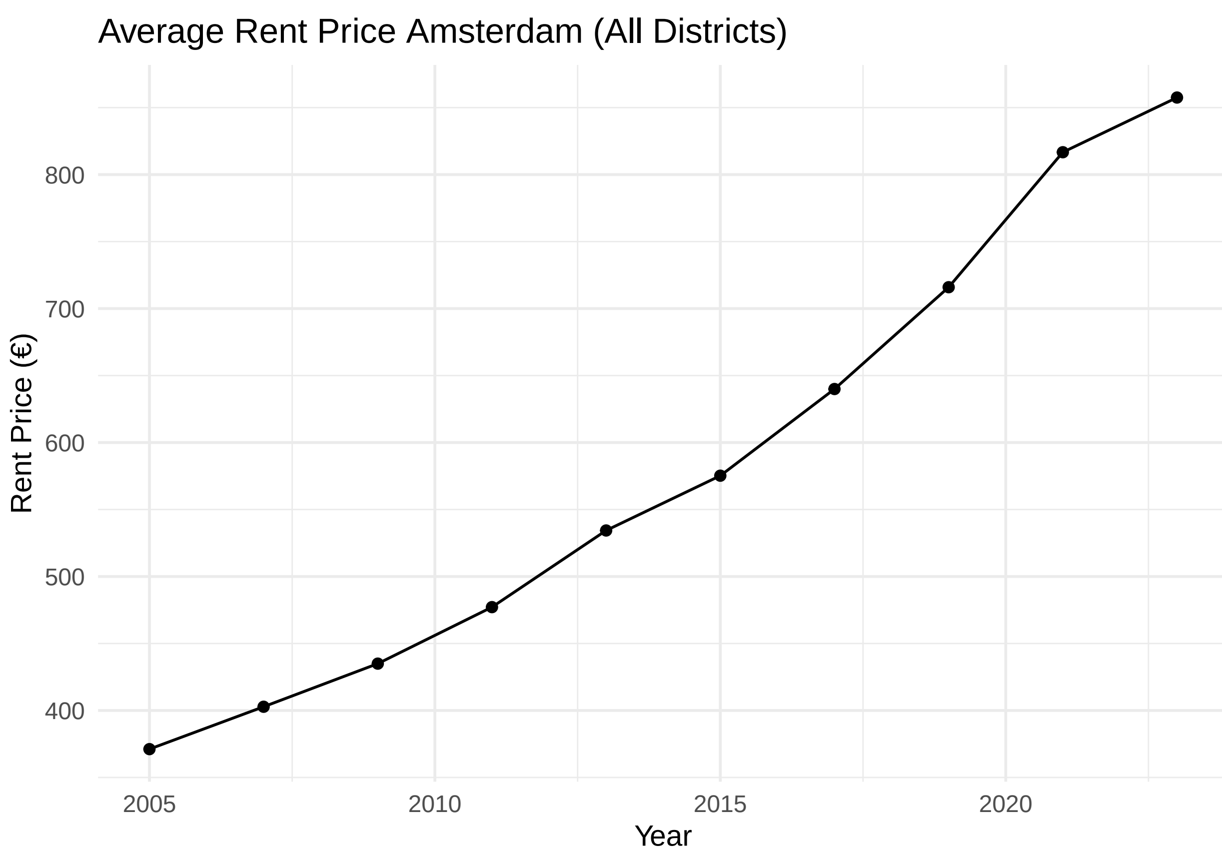
The variable **`perc_studenten`** was created by dividing the number of residents aged 15–25 by the total number of inhabitants in each neighbourhood and multiplying this by 100. This variable measures the percentage of young adults in a neighbourhood.

Second, we created **`rondkomen_cat`**. This variable groups the percentage of people who struggle to make ends meet into six categories. The categories range from **6.7% to 36.6%**. This makes it easier to show the data on a map.

Third, we created **`studenten_cat`**. This variable groups the percentage of young people into six categories, from **0% to above 20%**. This helps compare neighbourhoods more easily.

These new variables make the data easier to understand and easier to visualise. However, grouping values into categories means that some detailed information is lost.

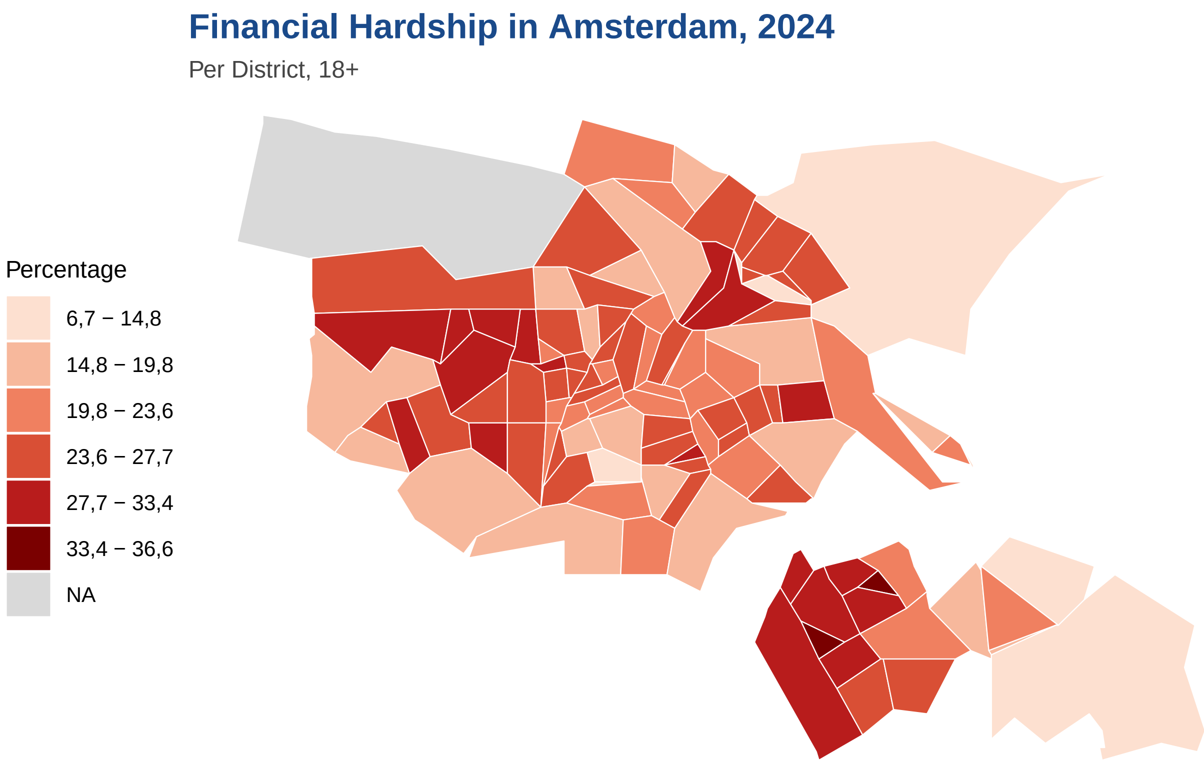
3.3 Visualize temporal variation



3.4 Visualize spatial variation

```
## Reading layer ‘wijk_2023’ from data source
##   ‘https://cartomap.github.io/nl/wgs84/wijk_2023.geojson’ using driver ‘GeoJSON’
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS:  WGS 84

## Rows: 91550 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (6): Marges, WijkenEnBuurten, Perioden, Gemeentenaam_1, SoortRegio_2, Co...
## dbl (3): ID, Leeftijd, MoeiteMetRondkomen_32
##
## i Use ‘spec()‘ to retrieve the full column specification for this data.
## i Specify the column types or set ‘show_col_types = FALSE‘ to quiet this message.
```



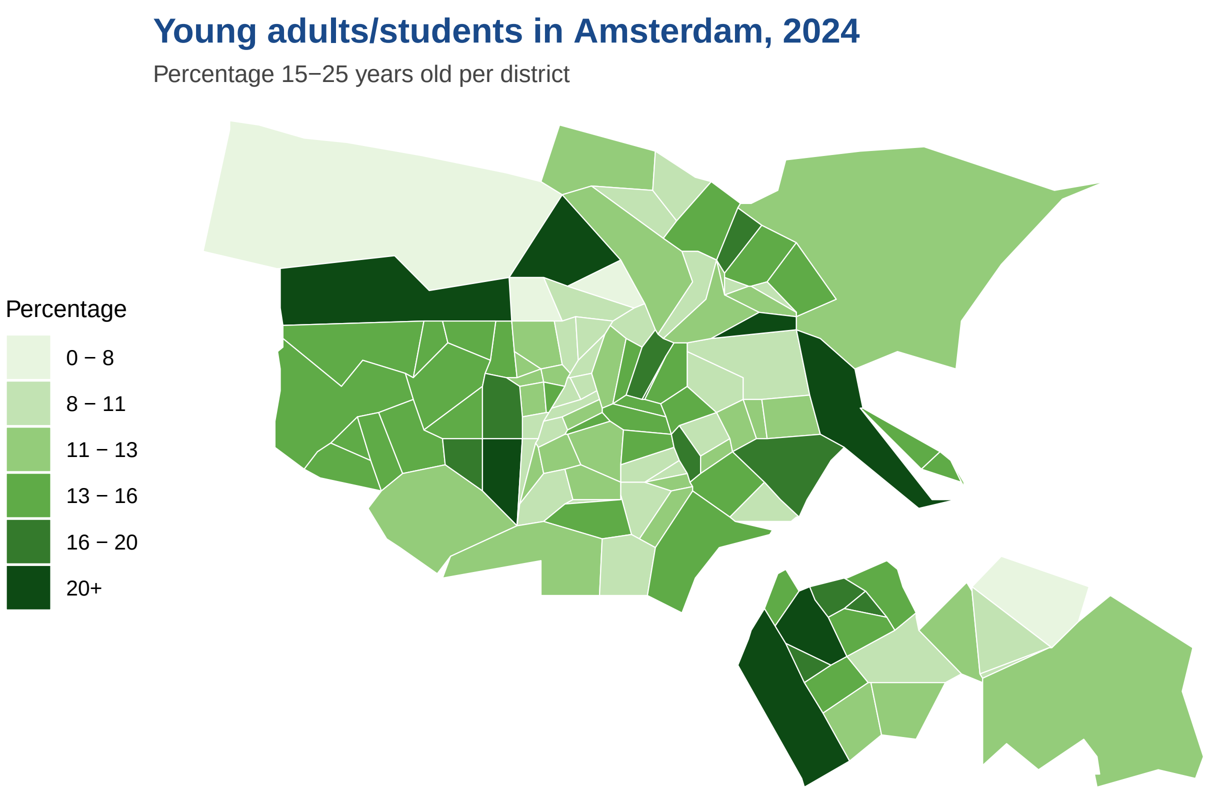
Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

```
## Reading layer ‘wijk_2023’ from data source
##   ‘https://cartomap.github.io/nl/wgs84/wijk_2023.geojson’ using driver ‘GeoJSON’
```

```
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS:  WGS 84
```

```
## Rows: 633 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (1): Wijken en buurten
## dbl (6): Bevolking/Aantal inwoners (aantal), Bevolking/Leeftijdsgroepen/0 to...
##
## i Use ‘spec()‘ to retrieve the full column specification for this data.
## i Specify the column types or set ‘show_col_types = FALSE‘ to quiet this message.
```



Analyze the relationship between two variables.
Here you provide a description of why the plot above is relevant to your specific social problem.

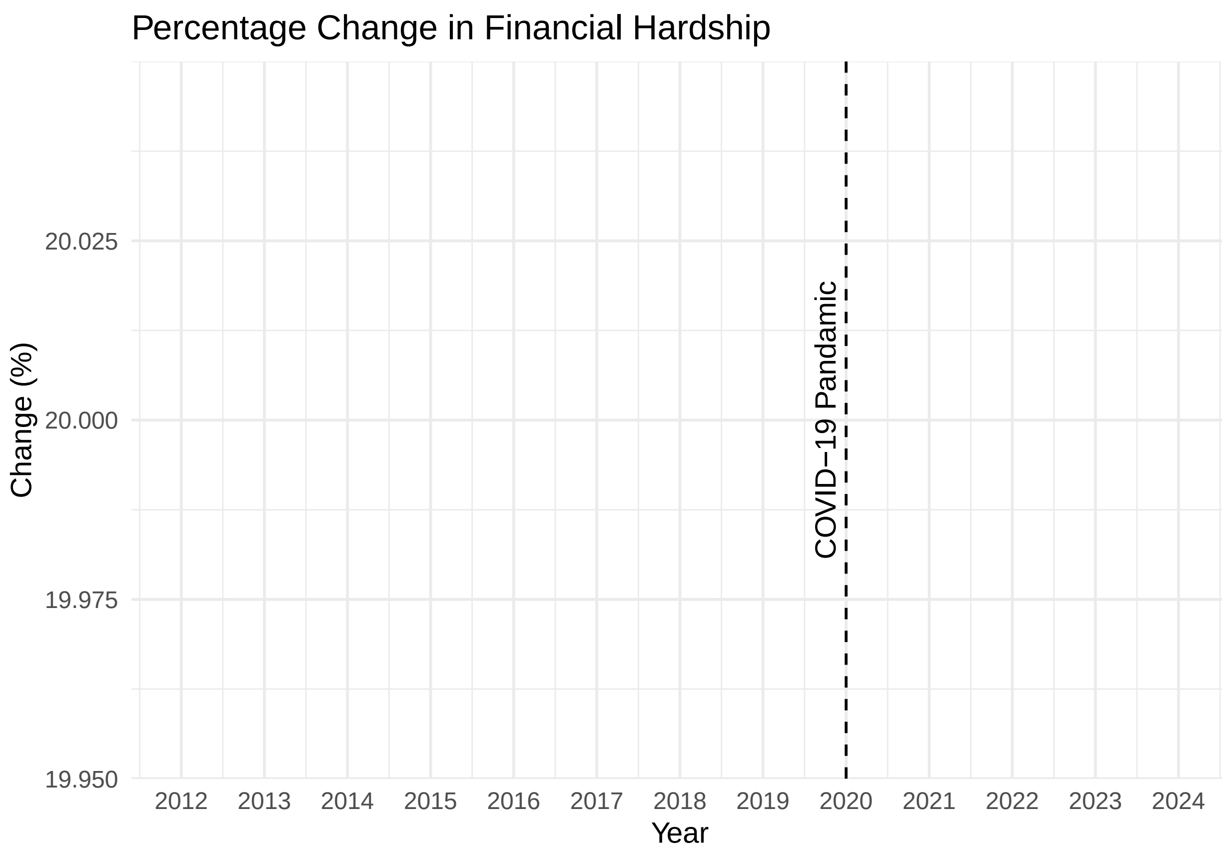
3.6 Event analysis

```
## Warning: There were 5 warnings in ‘summarise()‘.
## The first warning was:
## i In argument: ‘gem_moeite = mean(MoeiteMetRondkomen_32, na.rm = TRUE)‘.
## i In group 1: ‘Jaar = 2012‘.
```

```
## Caused by warning in ‘mean.default()‘:
## ! argument is not numeric or logical: returning NA
## i Run ‘dplyr::last_dplyr_warnings()‘ to see the 4 remaining warnings.

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (‘geom_line()‘).

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



Part 4 - Discussion

4.1 Discuss your findings

The findings indicate that financial hardship in Amsterdam is closely related to rising living costs and differences in income across neighbourhoods. The rental price data show a strong increase in average rents between 2005 and 2023, with average rents more than doubling during this period. This suggests that housing has become significantly less affordable, especially for households whose incomes have not increased at the same pace.

The income dataset reveals substantial differences in average household income between Amsterdam neighbourhoods. Some neighbourhoods have considerably higher average incomes than others, indicating that residents do not experience rising living costs in the same way. For households in lower-income neighbourhoods, increasing housing costs are likely to have a greater impact on their financial situation because a larger share of their income must be spent on essential expenses.

The dataset on difficulties making ends meet provides further evidence that financial vulnerability remains an important issue in Amsterdam. When viewed together, the income, housing, and financial hardship data suggest that residents with lower incomes are more likely to experience financial pressure. Rising rents, combined with increasing costs for food, energy, and other necessities, may leave these households with limited financial flexibility.

External factors should also be considered. The COVID-19 pandemic affected employment, income security, and economic activity, while the subsequent period of inflation increased the cost of living. These developments likely intensified financial difficulties for many residents and may help explain why a significant share of Amsterdam's population continues to struggle to make ends meet.

A limitation of this analysis is that it does not include all possible determinants of financial wellbeing, such as debt levels, savings, household composition, or educational attainment. Nevertheless, the combined evidence from the datasets suggests that rising housing costs and unequal income levels are key factors contributing to financial insecurity in Amsterdam.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/Wouter1334/programming4econ_groepje4

5.2 Reference list

Centraal Bureau voor de Statistiek. (2025, September 11). *Meer mensen tevreden over financiën in 2024, zorgen minder groot.* https://www.cbs.nl/nl-nl/nieuws/2025/37/meer-mensen-tevreden-over-financien-in-2024-zorgen-minder-groot

Mullainathan, S., & Shafir, E. (2013). *Scarcity: Why having too little means so much.* Times Books.

Buurtatlas. (z.d.). *Buurtatlas.* Volksgezondheidenzorg.info. Geraadpleegd op 22 juni 2026, van https://buurtatlas.vzinfo.nl